

BSc Information Technology · Year 3 Project

Project Proposal

AutoInsight ZA

Car Market Intelligence Platform

Data Engineering · Market Analytics · Web Platform · Power BI Visualisation

Allowing every South African to buy smarter.

March 2026

Richfield Institute

Project Snapshot

Project Name	AutoInsight ZA: South African Car Market Intelligence Platform
Qualification	BSc Information Technology: Year 3 Project
Core Software	Python (Playwright) · MySQL · HTML/CSS/JS · Power BI
Data Sources	Cars.co.za · AutoTrader.co.za · MotusSelect.co.za
Target Brands	Top 15 most-listed car brands in South Africa
Duration	February - May 2026 (approx. 4 months)

Problem Statement: Why This Project Matters

The average South African buying a second-hand car makes un-informed purchasing decisions.

With no single aggregated platform, buyers must manually browse multiple sites, guess whether a price is fair, and have no way to track how prices have moved over time. Car dealers are aware of this gap and price vehicles to take advantage of this.

South Africa's used-car market processes thousands of transactions per year. Prices are influenced by Rand exchange rate shifts, import duty changes, fuel cost pressures, and regional supply-demand shifts: yet consumers have access to none of this context at the point of purchase. AutoInsight ZA will be built to change that.

Without Our Project	With Our Project	Who Benefits
No price benchmarks : buyers rely on gut feelings	Instant fair-value score and pricing data for key makes/model/year	First-time car buyers
Cannot compare prices across multiple sites	Aggregated data from major SA portals	Budget-conscious families
No visibility of price history or trends	Track price inflation / deflation over time	Repeat buyers & upgraders
Provincial price differences not clear to buyer	Provincial heat maps revealing price variance	Out-of-province shoppers
Private vs. dealer pricing gap is not known	Clear comparison: private seller vs. dealership	Value-seekers
No forecast on future price movements	ML price predictions per vehicle type and brand	Long-term planners & investors

Project Structure



Layer 1: Gathering Data	Python + Playwright scrapes publicly listed vehicle data from Cars.co.za, AutoTrader, and Motus Select across the top 15 SA car brands. Fields captured include: Make, Model, Year, Mileage (KM), Price (ZAR), Province, Body Type, Transmission, Fuel Type, Seller Type, and meta data.
Layer 2: MySQL Database	A fully normalised MySQL schema will store all scraped data with dimension and fact tables (dim_make, dim_province, fact_listing, fact_listing_price_history, etc.). Creating tables, scrape-run logs, and price-history tracking allows for time-series and trend analysis from the start of the project.

Layer 3: Analytics & ML	Pandas and NumPy for data cleaning and transformation. A scikit-learn price forecasting model (Random Forest Regressor) is trained on listings to create a predicted market value for any vehicle. Model outputs are written back to the database to be shown on the front end.
Layer 4: Web Platform	A lightweight React (JavaScript) front-end site: built with HTML and CSS : will be the consumer-facing interface. Users can search and filter vehicles, view market summaries, and access the Price Estimator tool. The site will be designed to be clean, responsive, and accessible to all South Africans.
Layer 5: Power BI Dashboard	Power BI's dashboard visualisation is embedded directly into the React web platform through an iFrame element, allowing for Power BI's charting functions without re-building from scratch. Dashboard views include: market overview, price trend lines, provincial heat maps, mileage vs. price scatter plots, dealer vs. private-seller comparisons, and bargain-alert panels.

Technology Stack

Scraping	Python · Playwright · BeautifulSoup4
Database	MySQL 8.x · SQLAlchemy · MySQL Workbench
Processing	Pandas · NumPy · Scikit-learn
Front End	React (JS) · HTML5 · CSS3
BI / Data Viz	Power BI (iFramed into React site)
Dev Tools	PyCharm · Git / GitHub · python-dotenv

High-Level Timeline

Mid Feb	Scraper development & database schema
Late Feb	ETL pipeline & data validation
March	Full rollout & data gathering
April	Analytics layer & ML price model
March - April	React web platform development
April - May	Power BI dashboard & iFrame integration
May - June	Testing, documentation & submission

Implementation Languages & Frameworks

The project will use a multi-technology stack optimized for each architecture layer:

- Python 3.11+: Data scraping (Playwright, Requests), ETL processing (Pandas, NumPy), ML modelling (Scikit-learn), database interfacing (SQLAlchemy).
- SQL (MySQL 8.x): Relational database management, query optimization, stored procedures, and indexing.
- HTML5/CSS3/JavaScript: Front-end web development without framework dependencies, allowing for lightweight deployment and wide browser compatibility.
- DAX (Power BI): Dashboard measure calculations, data modeling, visualization configuration.

Development tools include PyCharm (Python IDE), MySQL Workbench (database administration), VS Code (web development), GitHub (version control), and python-dotenv (environment configuration management).

Software Development Lifecycle Model

This project follows an Agile development methodology with iterative 2-week sprints, allowing for quick prototyping, continuous feedback from all team members, and refining the project layers.

Agile Justification

The Agile model is best fit for this project due to:

- Evolving Requirements: Scraping targets and ML feature engineering benefit from iterative changes and refining based on early-stage data quality checks.
- Incremental Delivery: Each layer (scraping → database → ML → web → BI) can be developed, tested, and validated independently before integration, reducing risk.
- Team Member Collaboration: Regular sprint reviews allow course correction and priority adjustments aligned with academic timelines and due dates.
- Technical Uncertainty: ML model accuracy and web scraping site changes need validation across sprints, making the waterfall's model upfront planning not practical for this project.

Sprint Structure

Sprint	Duration	Key Deliverables
Sprint 1-2	Mid-Late Feb	Scraper development, database schema design, initial data collection
Sprint 3-4	Early-Mid Mar	Full brand rollout (15 brands), ETL pipeline, data validation & cleansing
Sprint 5-6	Late Mar-Early Apr	ML feature engineering, Random Forest model training, prediction API development
Sprint 7-8	Mid-Late Apr	Web platform (HTML/CSS/JS), search/filter interface, Price Estimator front-end
Sprint 9-10	Early-Mid May	Power BI dashboard design, iFrame integration, final testing & documentation

Each sprint concludes with a retrospective to evaluate velocity, address blockers (e.g., API rate limiting, data quality issues), and refine backlog priorities for subsequent iterations.

Technology Stack Summary

Scraping	Python · Playwright · Requests
Database	MySQL 8.x · SQLAlchemy · MySQL Workbench
Data Processing	Pandas · NumPy · Scikit-learn (Random Forest Regressor)
Web Development	HTML5 · CSS3 · JavaScript
Business Intelligence	Power BI Desktop (iFramed into web platform)
Development Tools	PyCharm · VS Code · Git/GitHub · python-dotenv

Key Output

- Functional web scraper targeting 15+ South African car brands across 3 major portals (Cars.co.za, AutoTrader, MotusSelect)
- Normalized MySQL database with 100,000+ vehicle listings and historical price-tracking capabilities
- Trained machine learning model achieving high accuracy on the price prediction validation set
- Consumer-facing web platform (HTML/CSS/JS) with search, filtering, and Price Estimator tool
- Embedded Power BI dashboard with 8+ analytical views (trends, heat maps, scatter plots, comparisons)
- Comprehensive project documentation: technical report, user guide, API documentation, GitHub repository